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Second Semester

B.Tech AI & Data Science

23MAT112 MATHEMATICS FOR INTELLIGENT SYSTEMS II

Duration: Two Hours | Maximum: 50 Marks

CO	Course Outcomes
CO01	Apply the fundamental concepts of linear algebra and calculus to solve canonical problems analytically and computationally
CO02	Apply the theory of optimization in solving real world problems
CO03	Apply the concept of probability and random variables in solving real world problems

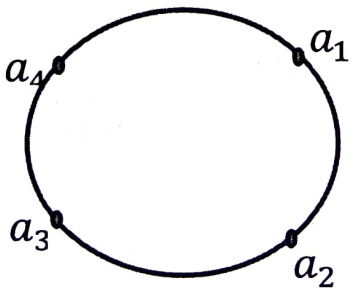
Answer all questions

1. ***Direct Approach for Data Reduction:*** **A direct approach of data reduction involves eliminating some portion in the data, uniformly across the data space, without losing critical information. Sometimes the elimination is done directly in the raw data, and other times, it is done after averaging the raw data over a neighbourhood.**

   1.1 Consider a discrete audio signal **X** of size 20×1. The objective is to perform data reduction by uniformly eliminating some of the samples from the raw data. Determine the transformation matrix **T** that will achieve a reduction factor of 5. **[3 Marks][CO01][BTL2]**

   1.2 Consider a discrete audio signal **X** of size 20×1. To achieve data reduction with better precision, we first compute the average of neighboring samples and then uniformly eliminate some of them. Determine the transformation matrix **T** that achieves a reduction factor of 5. **[4 Marks][CO01][BTL2]**

   1.3 Consider a cyclic data set, as illustrated in the figure below, where the data elements are positioned along a circle. **[7 Marks][CO01][BTL3]**



In a data reduction approach, the first step is to compute a weighted average of two neighboring elements in the clockwise direction using the equation:

$$B(i) = 0.8 \times A(i) + 0.2 \times A(i+1)$$

where **A** is the input data vector, and **B** is the resulting data vector after the averaging operation. Next, certain elements in **B** are eliminated uniformly across the data space, based on the desired reduction factor.

Let **C** be the reduced data set obtained after applying the aforementioned approach. Determine the transformation matrix **T** for the case where A is a 4×1 vector and the data reduction factor is 2, such that:

$$C = T \times A$$

2. Using any method of your choice, demonstrate that the data matrix $\begin{pmatrix} a & a & a \\ b & b & b \\ c & c & c \end{pmatrix}$ contains sparsity by representing it as a combination of sparse matrices. **[5 marks][CO01][BTL3]**

3. Consider the vector $A = \begin{pmatrix} 2a \\ 4a \\ -6a \end{pmatrix}$, where a is a non-zero real number ($a \neq 0$). Let the basis vectors be $v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, v_3 = \begin{pmatrix} -5 \\ 4 \\ 1 \end{pmatrix}$.

3.1 Which vector space is A a subset of? **[1 marks][CO01][BTL1]**

3.2 Is A a sparse vector? Give reasons. **[2 marks][CO01][BTL4]**

3.3 Prove that v_1, v_2 *and* v_3 form a set of basis vectors, with reasons. **[3 marks][CO01][BTL4]**

3.4 Prove that A contains 66% sparsity when spanned across the basis vector space of v_1, v_2 *and* v_3. **[7 marks][CO01][BTL5]**

4. **<u>Singular Value Decomposition (SVD):</u> SVD expresses any $m \times n$ matrix A as a product of three matrices, $A = U\Sigma V^T$.**

4.1 Determine the sizes of the matrices U, Σ *and* V for the SVD decomposition of any data matrix A of size 1000×45. **[2 marks][CO01][BTL4]**

4.2 Can the matrix

$$\Sigma = \begin{pmatrix} 0.5 & 0 & 1 \\ 0 & 10.67 & 2 \end{pmatrix}$$

represent the Σ matrix in an SVD decomposition. Provide reasons for your answer. **[3 marks][CO01][BTL4]**

4.3 Can the matrix

$$U = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

represent the U matrix in an SVD decomposition. Provide reasons for your answer. **[3 marks][CO01][BTL4]**

4.4 Consider a 2-dimensional symmetric data matrix $A = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$. Determine the SVD decomposition $(\boldsymbol{U, \Sigma \text{ and } V})$ of the data matrix A, valid for all values of a and b.

[10 marks][CO01][BTL5]

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO01	**50**	BTL 1	**1**
CO02	**0**	BTL 2	**7**
CO03	**0**	BTL 3	**12**
		BTL 4	**13**
		BTL 5	**17**
		BTL 6	**0**